New-2

MEMBERSHIP

Stars in catalog	90
with Gaia GSP-Phot	52 (58%)
with radial velocity	18
Stream id (sID)	8

GEOMETRY · great-circle fit

Centroid RA, Dec	38.2, 4.3 deg
Centroid l, b	162.3, -49.3 deg
Pole RA, Dec	211.7, 85.3 deg
Length (φ1, 2–98%)	46.8 deg
Width σ(φ2), robust	4.52 deg

KINEMATICS

pmRA* med ± σ	-9.48 ± 11.36 mas/yr
pmDE med ± σ	-39.97 ± 9.36 mas/yr
μφ1* med ± σ	-9.18 ± 10.43 mas/yr
μφ2 med ± σ	-40.21 ± 9.55 mas/yr
Parallax (cat.) med	0.877 ± 0.277 mas
Parallax (wtd, ZP-corr.)	0.871 ± 0.014 mas
Distance	d = 1.1 ± 0.0 kpc
Physical length	0.94 kpc
RV median ± σ(MAD)	182.1 ± 12.7 km/s
RV gradient dV/dφ1	0.57 km/s/deg

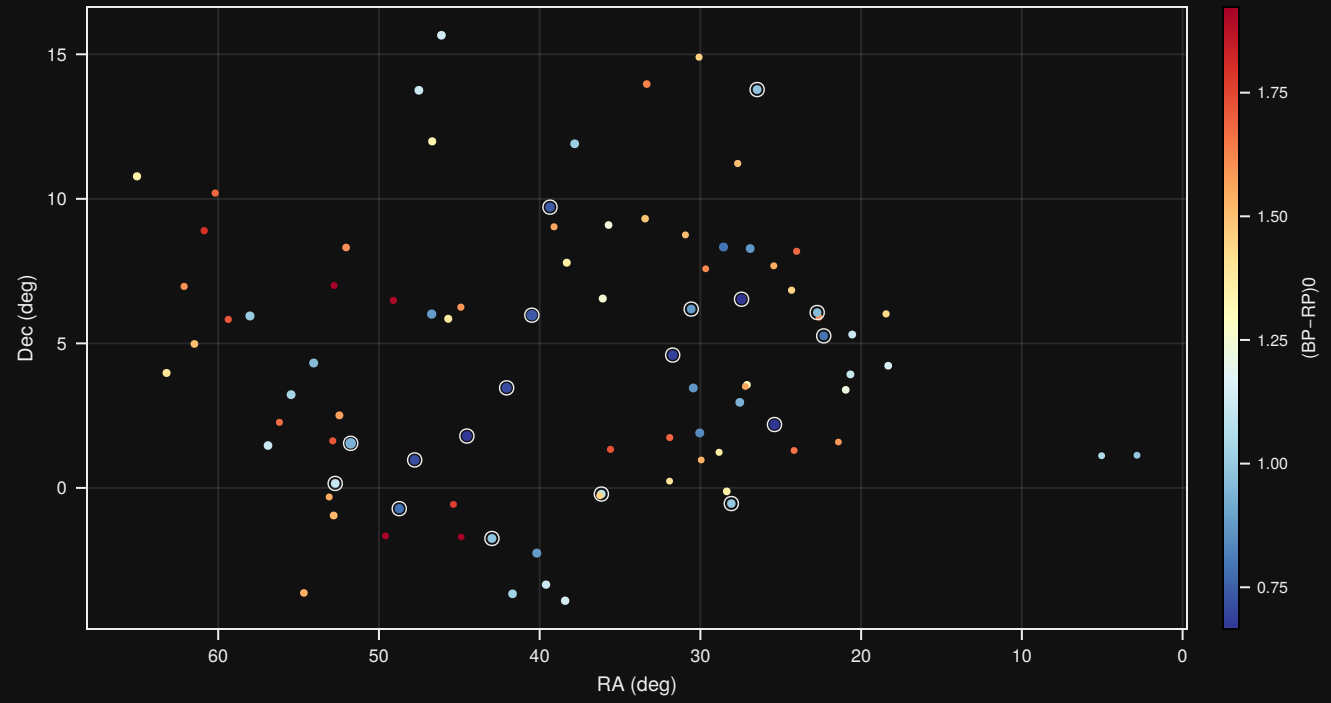
PHOTOMETRY · POPULATIONS

G0 range (median)	11.6–20.0 (18.3)
(BP−RP)0 median	1.34
GSP-Phot [M/H] (n=52)	-1.86 ± 0.53
GSP-Phot Teff median	4900 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (42 stars)

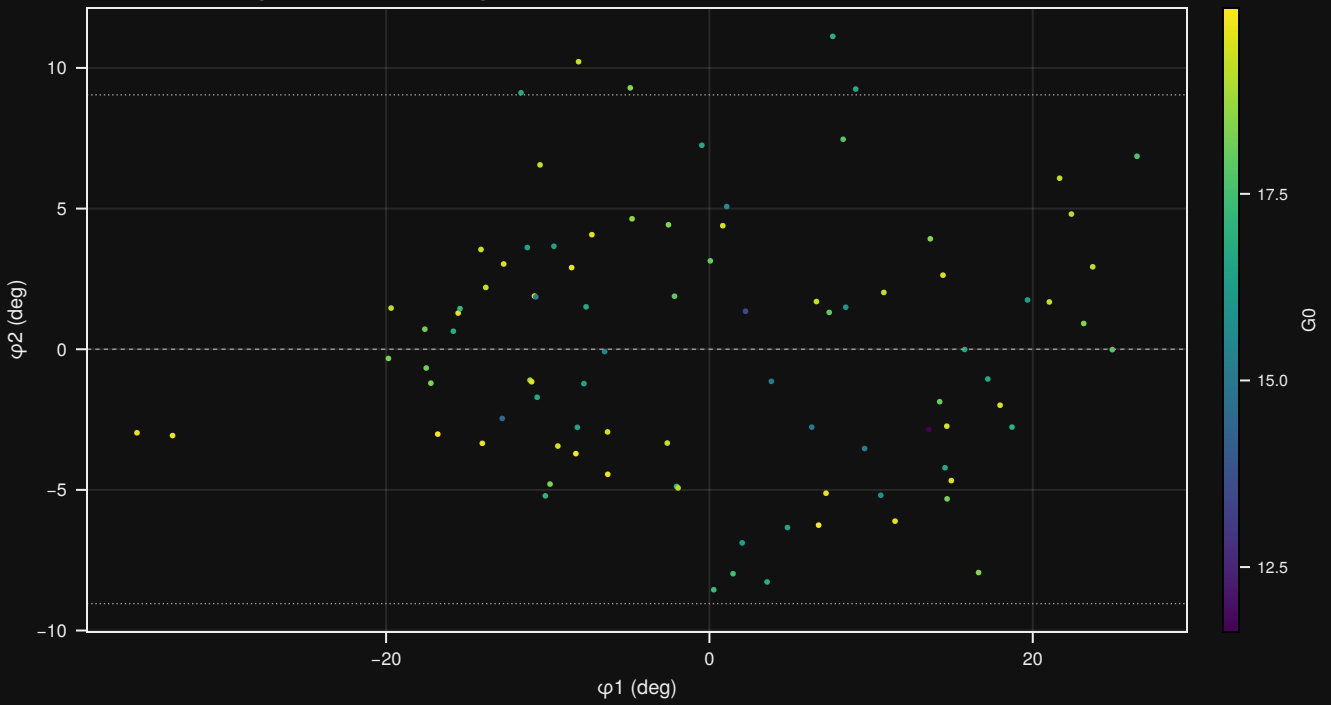
IBATA+2024 · TABLE 3

n catalog | paper	90 | 90
n_RV catalog | paper	18 | 18
<[Fe/H]> paper	-2.07
Stellar mass (paper)	73 Msun
Pole offset vs paper	1.8 deg

Sky map · catalog members (n = 90) · size ∝ brightness · ring = has RV

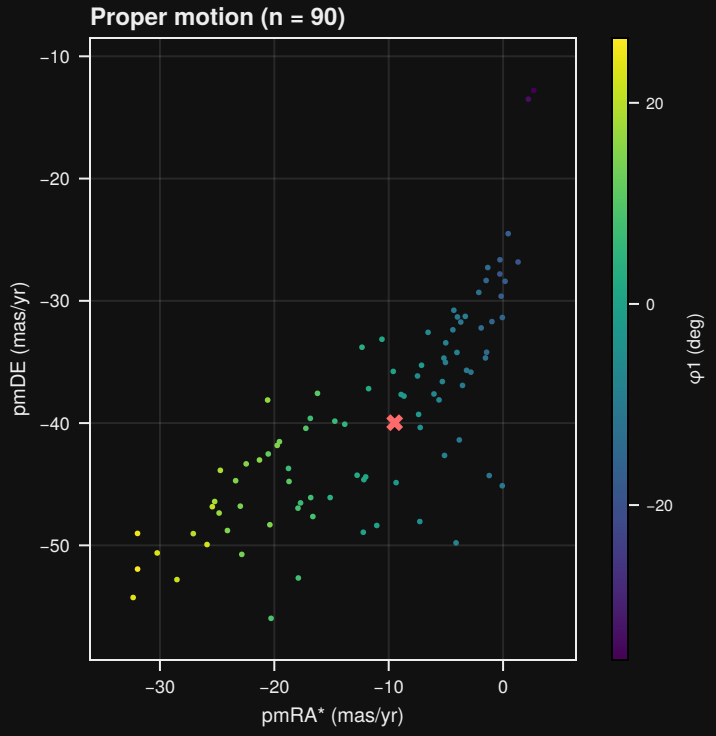


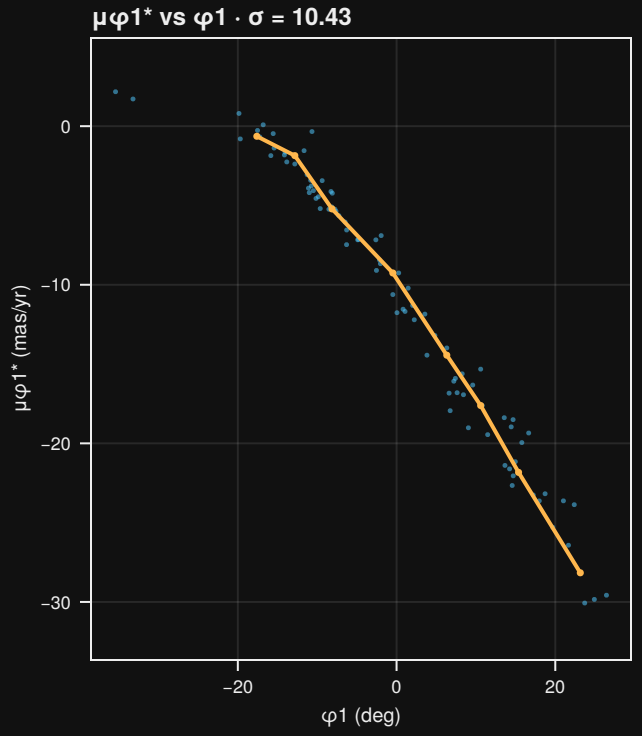
Stream frame (great-circle fit) · length 46.8° · width σ 4.52°

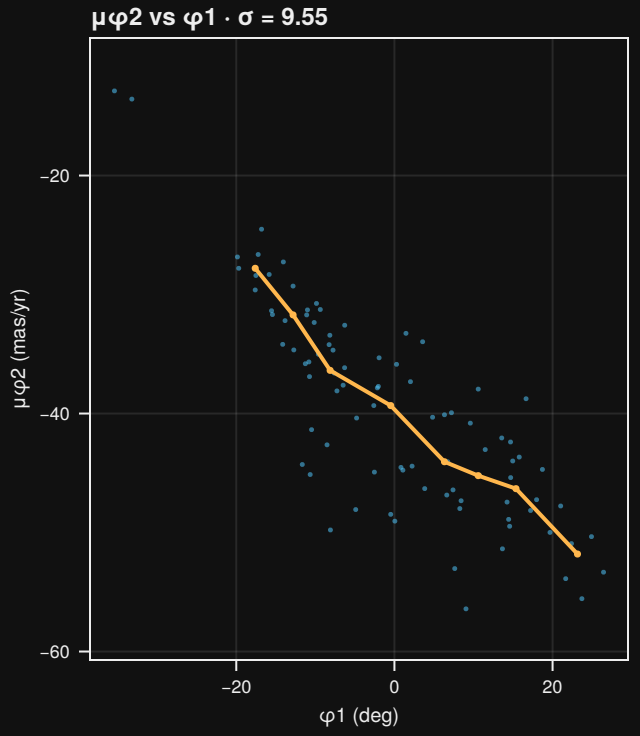


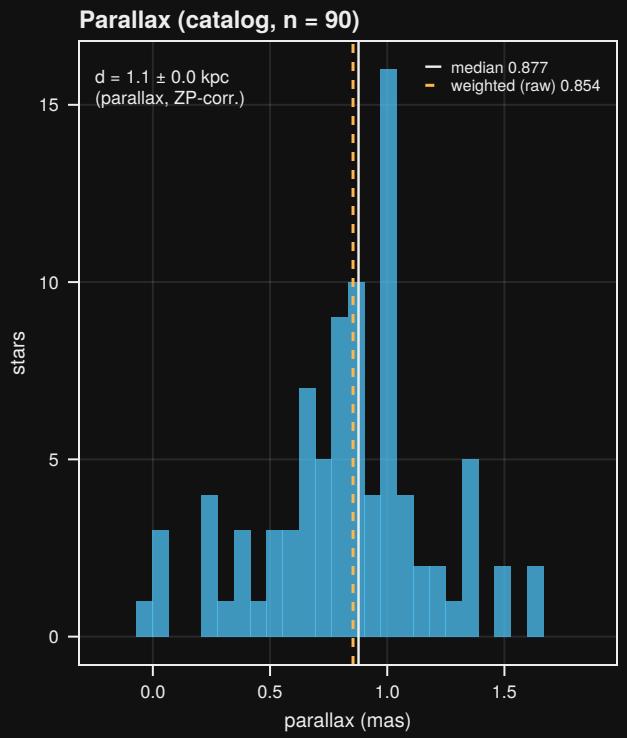
New-2

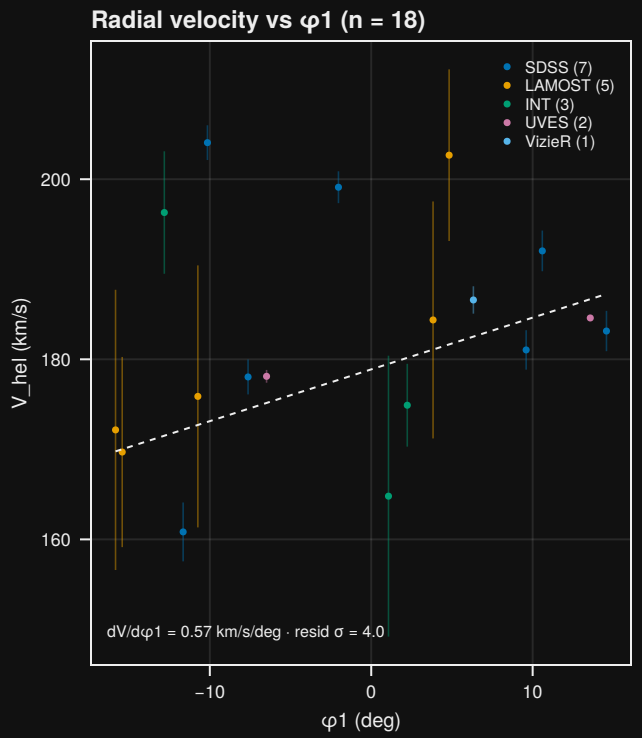
Kinematics · proper motion · parallax · radial velocity

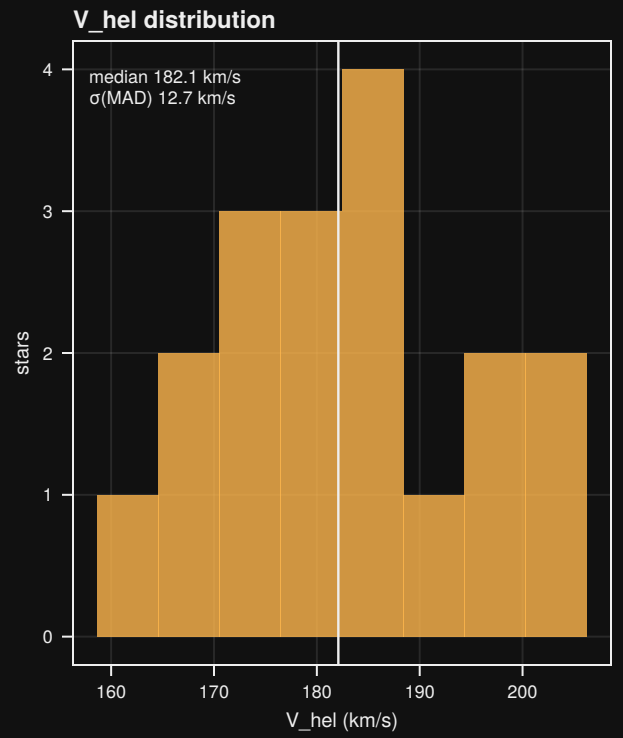








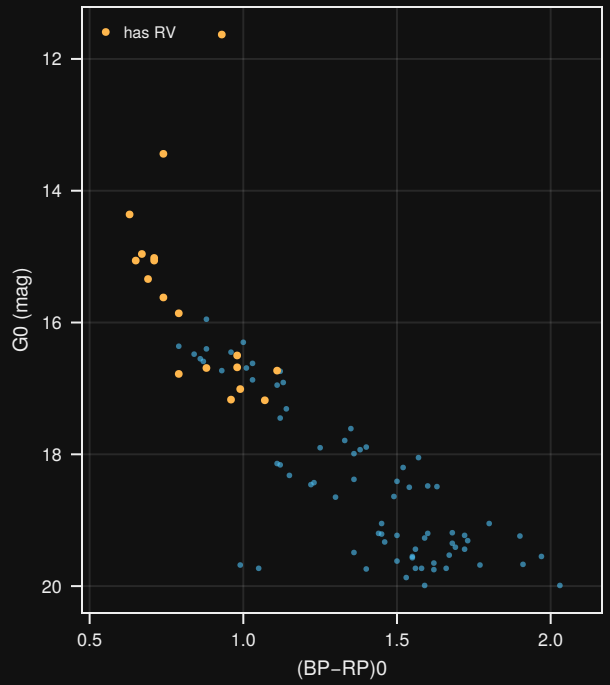




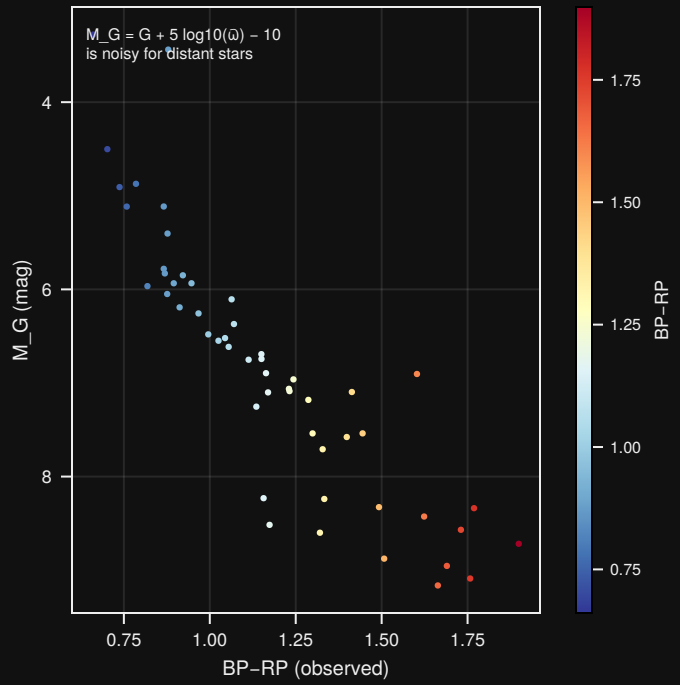
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Photometry · stellar types

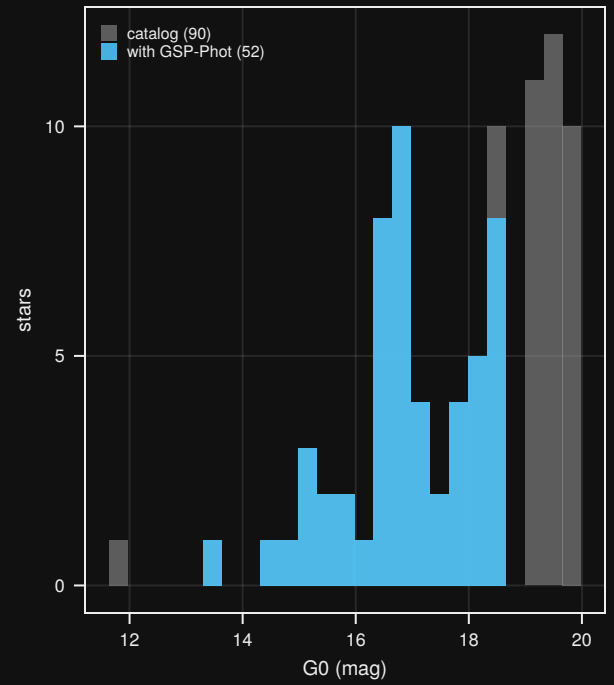
CMD · dereddened (n = 90)



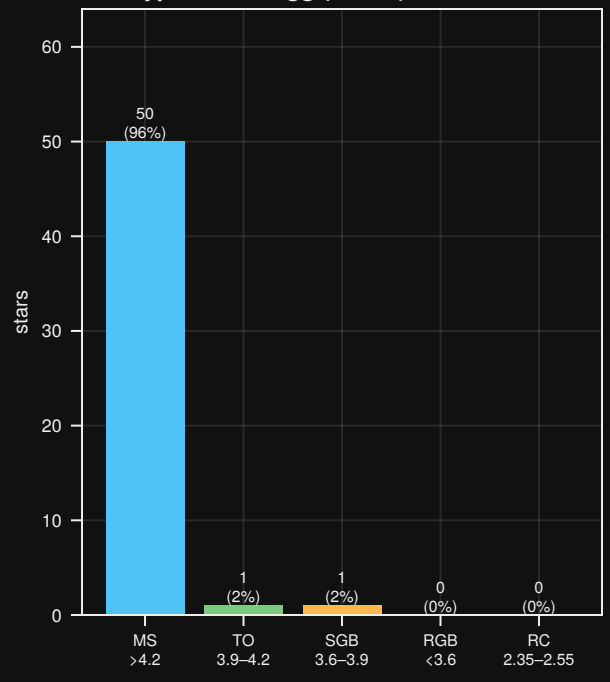
HR diagram · ϖ>0 (n = 52/52)



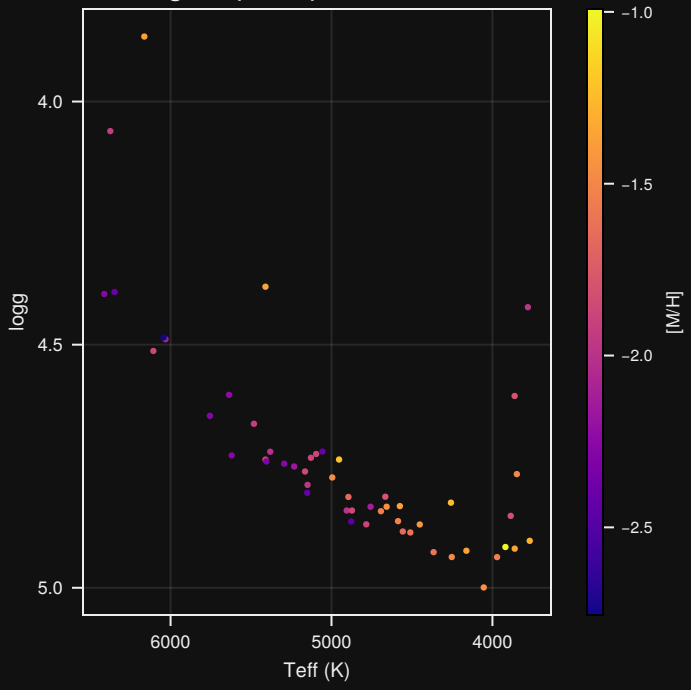
G0 & GSP-Phot coverage



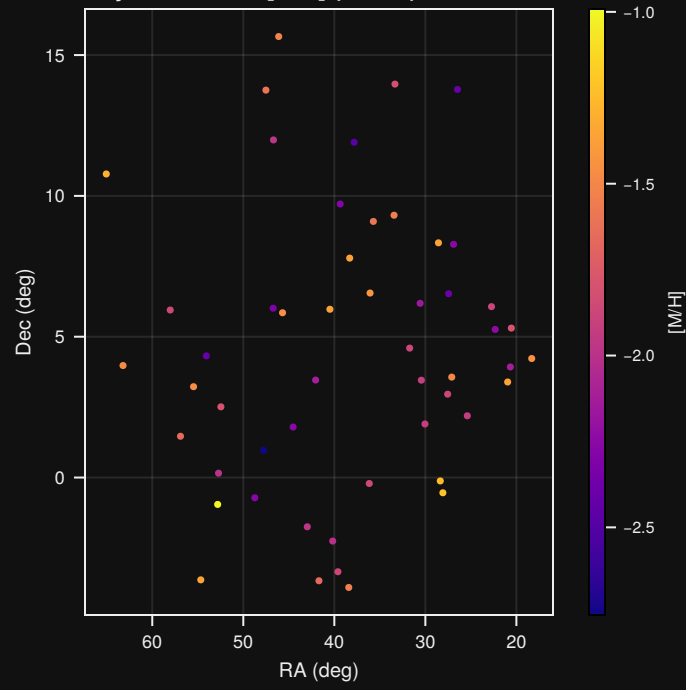
Star types from logg (n = 52)



Kiel diagram (n = 52)



Sky · GSP-Phot [M/H] (n = 52)

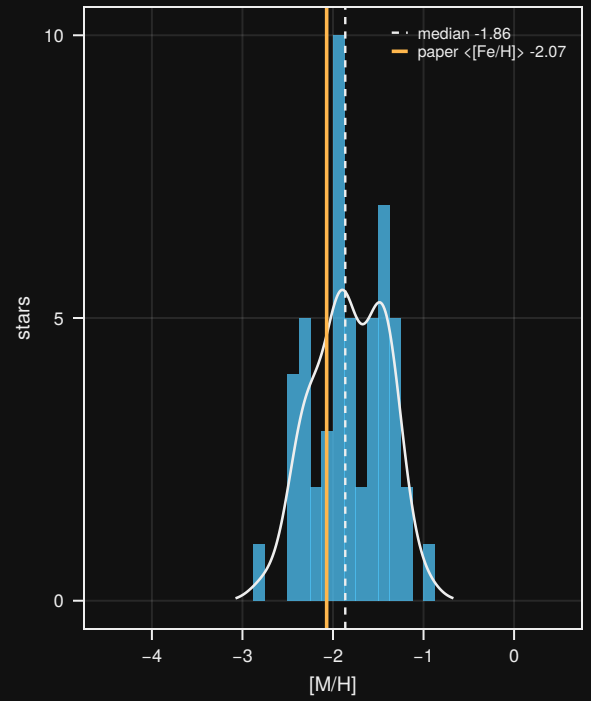


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

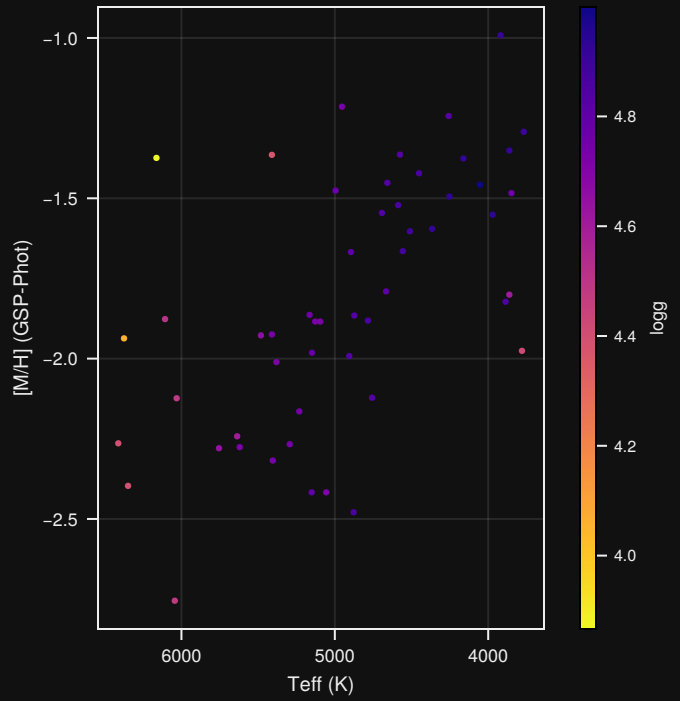
New-2

Metallicity · age · astrometric quality

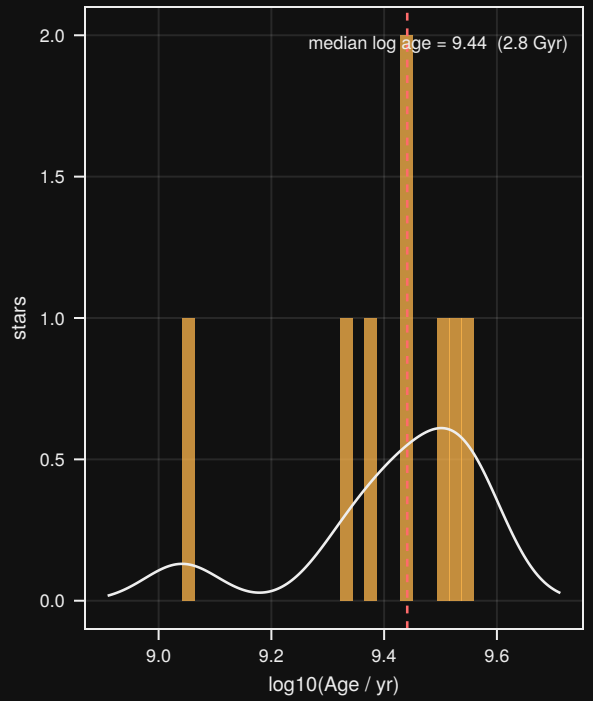
[M/H] GSP-Phot (n = 52)



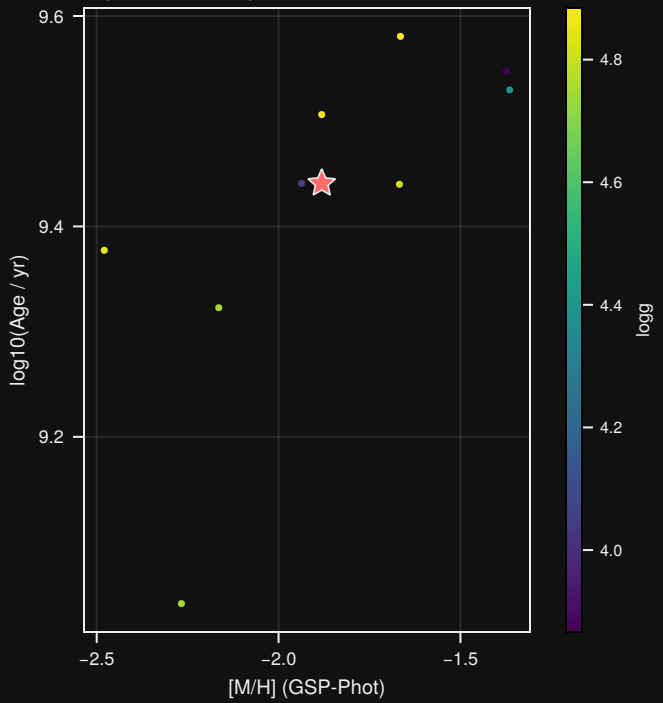
Teff vs [M/H] (n = 52)



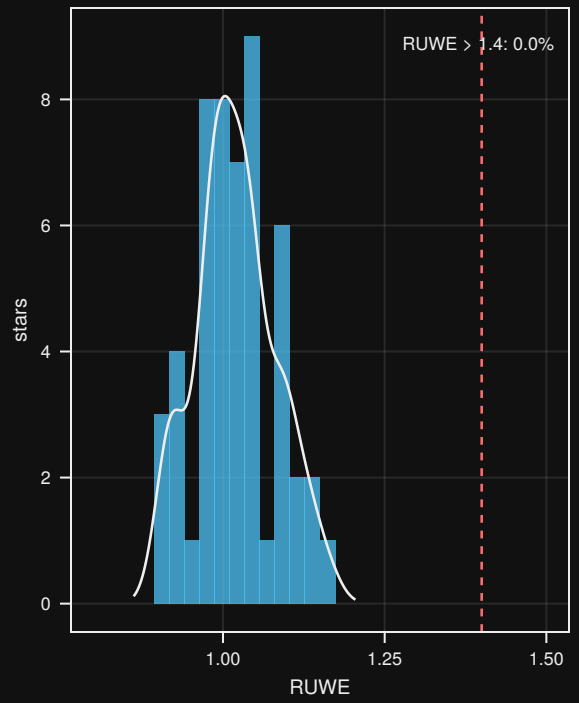
FLAME age (n = 9)



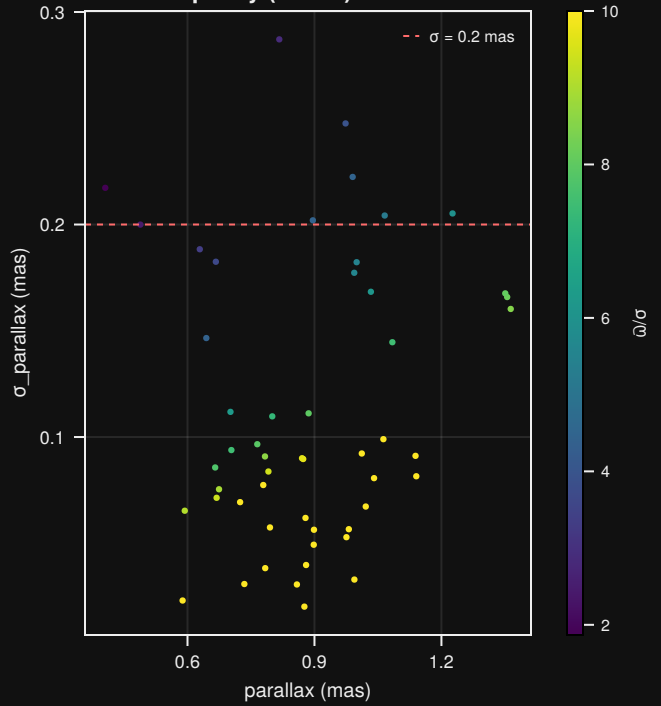
Age–metallicity (n = 9)



RUWE (n = 52)

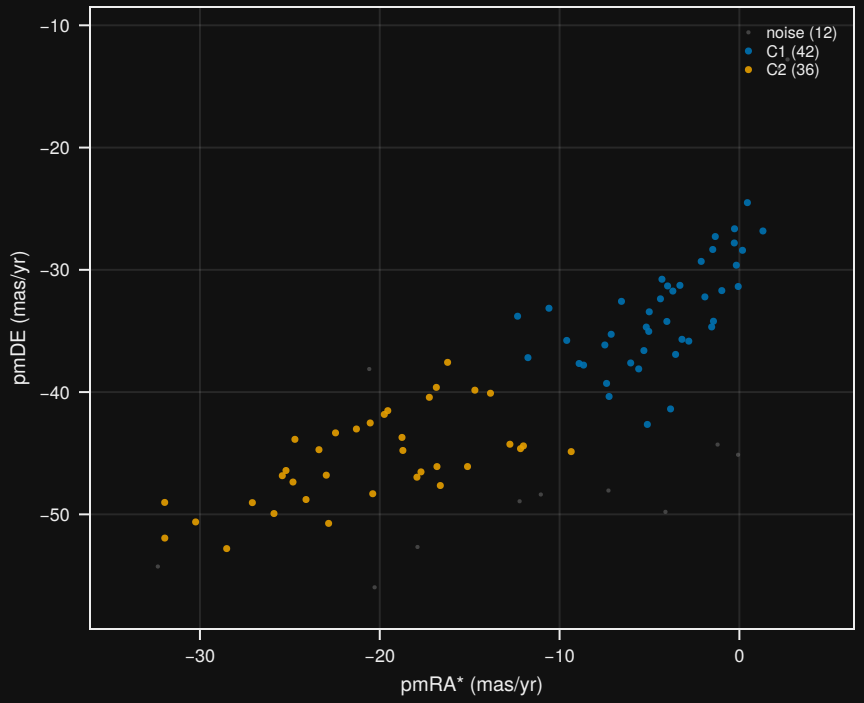


Parallax quality (n = 52)

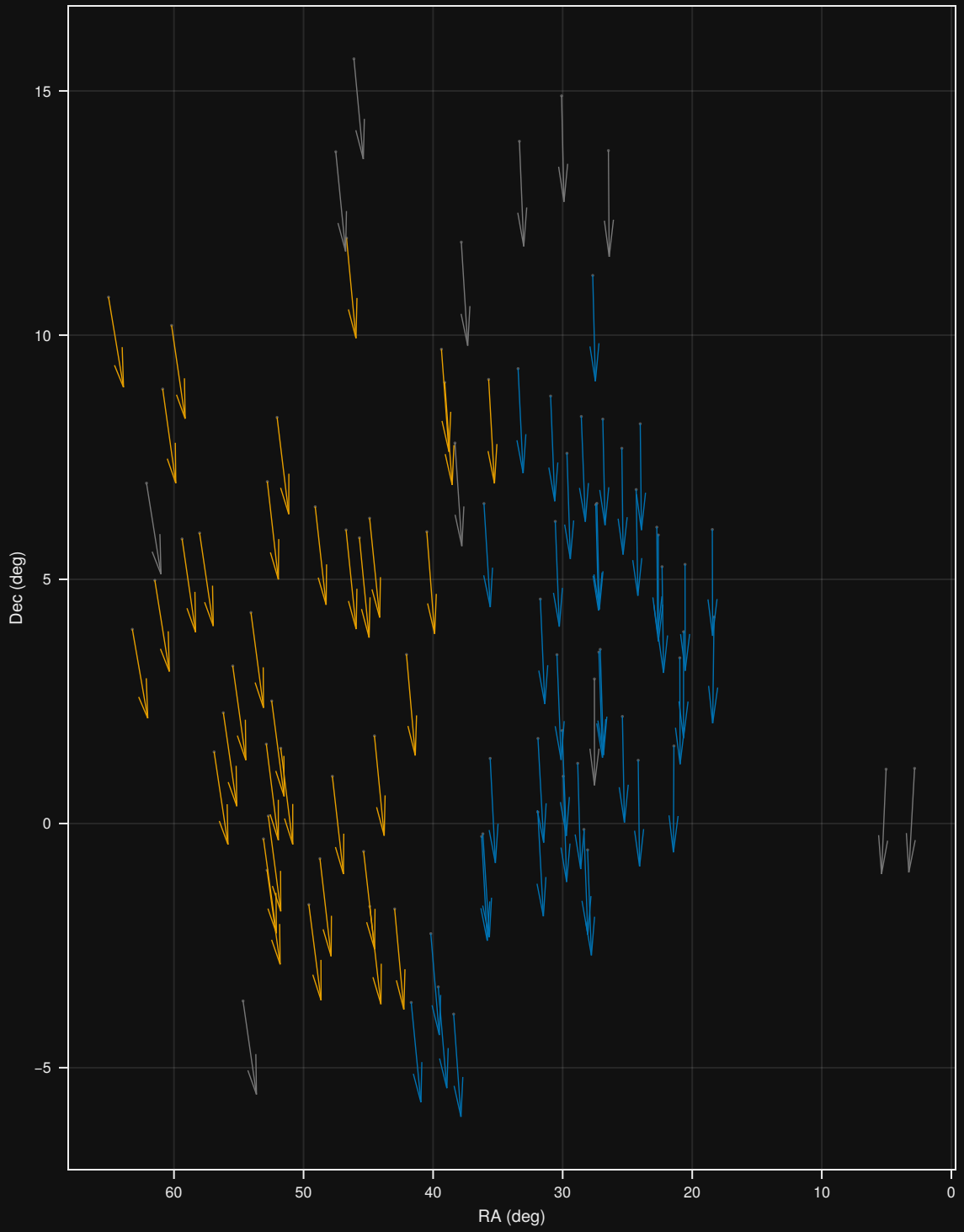


New-2

PM clustering · HDBSCAN · 2 cluster(s) · noise 13%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	12	-9.16	-48.21	13.22	-1.71	1
C1	42	-4.01	-34.00	3.76	-1.87	8
C2	36	-20.07	-45.48	5.84	-1.92	9

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Per-cluster proper-motion vector maps

